MIRAZ HOSSAIN

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Education

• Ahsanullah University of Science and Technology (AUST) Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering (EEE); CGPA: 3.025/4.00 Nov 2021 – Nov 2025
Relevant Courses: *Circuit Design, Power System, Control System Design, MCU, Computer Architecture, Digital Signal Processing, Optical Communication, C++ Programming, VLSI, Image Processing, Medical Device Design, CAD.*

Work Experience

• Frontier Semiconductor (FSM) Dhaka, Bangladesh
Embedded Systems Engineer May 2026 – Present
- Develop production C/C++ firmware for semiconductor metrology and wafer inspection equipment on STM32 and Raspberry Pi Compute Module platforms.
- Implement closed-loop stepper and servo motion control for multi-axis stages, with hardware fault handling and state persistence in Flash memory.
- Design digital filtering algorithms to smooth optical sensor data for high-precision wafer measurement.
- Build Python desktop diagnostic and production GUIs for automated hardware verification.
- Integrate industrial machine vision cameras with OpenCV-based image processing for automated wafer alignment; maintain team Git firmware workflow.

• Cybernetics Hi-Tech Solution (Pvt.) Ltd. Dhaka, Bangladesh
Embedded Systems Engineer – R&D Intern Feb 2026 – Apr 2026
- Programmed ARM Cortex-M0 firmware for smart utility meters: anti-theft protection, 20-digit STS token cryptography, DLMS communication and PLC module calibration; studied DCU/AMI metering communication.
- Integrated ROS and Micro-ROS with microcontrollers for real-time navigation and control in Autonomous Guided Vehicles (AGVs) for agricultural and factory use.
- Developed Raspberry Pi and camera-based object tracking for drone navigation.
- Verified team hardware/firmware design decisions against industry standards.

Publications

- R. Yeassin, T. S. Saiket, F. H. Fahad, A. Mia, M. Tarekuzzaman, M. E. Hoque, M. M. R. Adnan, **Miraz Hossain**, S. Chakraborty, M. A. Sufian, M. U. Koraishe, and M. Hasan, *"Hydrogen economy pathways for net-zero transitions: System integration, scalability barriers, and strategic deployment,"* **Energy Conversion and Management: X** (Elsevier, Q1), Vol. 31, Art. 102165, July 2026.
  - **Key Focus:** Synthesized system-level integration, techno-economic feasibility, grid/storage balancing, and scalability barriers across green and low-carbon hydrogen pathways for industrial decarbonization.
  - **Links:** DOI: https://doi.org/10.1016/j.ecmx.2026.102165 | ScienceDirect: https://www.sciencedirect.com/science/article/pii/S2590174526006483

Thesis

- **AI Adaptive PID Controller –** Designed an adaptive AI-based PID controller for multi-axis robotic arm motion control.

Management Experience

- **AUST Innovation and Design Club (IDC) –** *Executive Member & Assistant Head of Project Team*
- **AUST Satellite Lab –** *Founding Member*
- **AUST Model Rocketry –** *Software and Navigation Team Member*

Logistics Experience

- **Sourcing:** Local component markets in Dhaka; global sourcing via Digikey, JLCPCB, Alibaba, Made-in-China, AliExpress; price negotiation on Chinese platforms.
- **Finance:** Compiled R&D and product development budgets.

Undergrad Projects

- **Computer Vision Gesture-Controlled Lighting:** Designed a custom PCB and OpenCV/MediaPipe gesture pipeline for touchless HMI load control.
- **16-DOF Humanoid Robot & Android Interface:** Designed laser-cut frame in AutoCAD; built Android app for real-time kinematic control and servo synchronization.
- **Overcurrent Protection & GSM Diagnostics:** Developed power protection system with GSM fault reporting and automated diagnostic alerts.
- **Vocal Navigation & Speech-Triggered Vehicle:** Built voice-actuated mobile robot using NLP to map speech commands to hardware actions.
- **IoT Health Monitoring & Personal Safety System:** Integrated GPS, environmental sensors and health telemetry over IoT for hazardous environments.
- **Self-Balancing Breadboard Platform:** Implemented Kalman filter sensor fusion and PID control for real-time stabilization.

- **Distributed Smart Home Framework:** Built appliance automation using Node-RED, ESP RainMaker and SinricPro.
- **Precision Agricultural Climate Optimizer:** Built multi-sensor IoT node with feedback loops to monitor and adjust farm micro-climate.
- **Spectral Signal Analysis:** Estimated speech signal spectra using Burg, Yule-Walker and Levinson-Durbin methods.
- **RFID Access & Attendance System:** Designed RFID-based attendance logging and encrypted entry management.
- **Autonomous Navigation Suite:** Built gyro-controlled, human-following, line-following and obstacle-avoiding robots using ultrasonic and IR arrays.
- **Industrial Smart Color Classification:** Built sorting system with high-speed color sensors and logic-based actuation.
- **NTP & RTC Smart Digital Clock:** Built ESP8266/MAX7219 clock synchronized over NTP, with 3D-printed enclosure.
- **IoT Smart Waste Management:** Built smart dustbin with ultrasonic fill-level and dust/air quality sensing, logged to ThingSpeak.
- **CanSat Weather Monitoring System:** Designed CanSat telemetry for atmospheric data acquisition and transmission.
- **Digital AC Voltage & Frequency Meter:** Built meter to monitor mains voltage and frequency fluctuation.
- **Robotic Arm IR-Control:** Developed multi-axis arm with IR remote control and programmed sequences.
- **Interactive Welcoming Robot:** Built talking robot with ID-card verification for entry.
- **Python Geospatial Geocoding App:** Created app to extract location data and coordinates from phone numbers.
- **ESP32 Air Mouse (HID):** Built wearable HID using ESP32 and IMU for gesture-based cursor control.
- **ML Diabetes Predictive Model:** Trained TensorFlow neural network for binary diabetes risk classification.
- **45nm CMOS Full Adder (VLSI):** Analyzed and optimized delay, power, PDP and EDP in Cadence Virtuoso.
- **Automated Fluid Management System:** Implemented sensor-driven water level control with alarm and pump automation.
- **Industrial Door Lock:** Designed multi-feature digital lock with encryption-focused security.

Skills Summary

- **Single Board Computer:** Raspberry Pi Compute Module, 4B, Pico
- **Microcontrollers:** STM32 (F4, L4, H7), ESP32, ESP32-C3, ESP32-C6, ESP8266, Arduino Mega, ARM Cortex-M0, HT5023, AT89C4051
- **Firmware & Control:** Closed-loop stepper/servo motion control, digital filtering (Moving Average, EMA, Kalman), sensor calibration, hardware fault handling, Flash state persistence, serial communication, machine vision (OpenCV), Python desktop tool/GUI development (CustomTkinter, Tkinter, PyInstaller)
- **Electrical Design:** Altium Designer, EasyEDA, Cadence Virtuoso, Proteus, Quartus Prime, Simulink, OrCAD PSpice
- **Protocols:** UART, RS-232, RS-485, SPI, I²C, CAN, Modbus, USB (CDC), Bluetooth, Wi-Fi, Ethernet, GSM/GPRS (AT Commands), GPS (NMEA), LoRa, MQTT, HTTP/HTTPS, TCP/IP, WebSockets, DLMS
- **Software and Tools:** C/C++, Python, Verilog, Git/GitHub, MATLAB and Simulink, Arduino IDE, STM32CubeIDE, Keil IDE, J-Link, VS Code, PlatformIO, MicroPython, OpenCV, MediaPipe, NumPy, TensorFlow, YOLO, DeepSORT, ROS/Micro-ROS, Linux (lgpio), Android Studio, MIT App Inventor, Thonny
- **Hardware and Tools:** Sensors & motion (BMP180/280, MPU6050, ADXL335/345, BNO055, GP2Y0A21YK0F, GP2Y1010AU0F, TCRT5000, TTP223, SW-420, TCS230/TCS3200, MQ-4, temperature sensors); positioning (NEO-6M/7M GPS, EC11 encoder); wireless (SX1278 LoRa, NRF24L01+, HC-05/06, SIM808, SIM800L, SIM900A, A7670C, MNL80A PLC, ASK 433 MHz, TSOP1738); programmers (FT232RL, ST-Link V2, USBasp); motion & drives (NEMA-17, 28BYJ-48, MG996R, BLDC/ESC, industrial servo/stepper drives, TB6600, L293D, L298N, TB6612FNG, BTS7960, Cytron MD10C, PCA9685); power (DALY BMS, LM2596, MT3608, LDO, ZMPT101B, ACS712, PZEM-004T, PCF8574); measurement (YF-S201, HX711, MAX30100, FC-28, HC-SR501); displays & HMI (TM1637, 16×2 LCD, SSD1306, MAX7219, ST7789V TFT, 4×4 keypad); storage & ID (DS3231, microSD, MFRC522); vision & systems (industrial machine vision cameras, ESP32-CAM OV2640, Pixhawk PX4, FlySky FS-i6X, PAM8610).
- **CAD Software:** SolidWorks, AutoCAD
- **Documentation Tools:** Microsoft Word, Excel, PowerPoint

Honors and Awards

- Invitation to Cozmoclench Robotics Competition, Techfest IIT Bombay – 2022

References

Mr. Muhammad Jakaria Rahimi
Associate Professor (Dept. of EEE), Ahsanullah University of Science and Technology (AUST)
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